CSE221

Fullmark: 15 Quiz no: 1 Summer2024

ID: _____ Name:_____ Time: 30 minutes

1. Given a program, write down the time complexity. (4 marks)

```
for ( i = 0; i<=n; i+=8){
        print("My Soldiers, Rage!");
        for(j=0; j<m; j+=2){
              if(j > 10){
                    break;
               }
        }
}
here, 10<m<10^10
```

2. Given a program, write down the time complexity. (4 marks)

```
for( i = n; i >= 1; i = i / 20 ){
        for( j = 1; j <= n; j = j - 3 ) {
                for( k=1; k<=40 ; k=k+1){
                        print("My Soldiers, Scream! ");
                }
         }
}
```

3. Given a program, write down the time complexity. (4 marks)

```
for( i = 1; i <= n; i = i * 20 ){
        for( j = 10; j <= n; j = j + 9 ) {
                for( m=1 ; m*m<n ; m=m*5 ){
                        print("My Soldiers, Fight!");
                  }

         }
}
```

4. Given a program, write down the time complexity. (4 marks)

```
for( p = n; p > 1; p = p/n ){
        for( j = 1; j*j <= n; j = j + 9 ) {
                for( m=1 ; z<=n ; m=m*3 ){
                        z = z+m
                }

        }
        for( k=1 ; k<l ; k++ ){
                print("Tatakae!");
        }
}
```

5. Write a correct asymptotic exact upper bound for any 2 of the below with proper calculation. [2 marks]

   a. f1(n) = 500n + 100$n^{1.5}$ + 50$nlog_{10}n$

   b. f2(n) = $(logn)^{2024}$

   c. f3(n) = $n * \sqrt[3]{n^2}$

   d. f4(n) = $0.003log_4 n + log_2 log_2 n$